PROJECT REPORT

ON

IDENTIFYING CROSS-SELL OPPORTUNITIES IN RETAIL BANKING THROUGH CUSTOMER PRODUCT PENETRATION DATA ANALYSIS

Submitted in partial fulfillment of requirements for the award of the degree of

Master in Business Administration

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DECLARATION

I hereby declare that the project report titled "Identifying Cross-Sell Opportunities in Retail Banking Through Customer Product Penetration Data Analysis" submitted for the Master in Business Administration is my original work and the project report has not formed the basis for award of any degree, diploma, associate ship or other similar titles.

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ABSTRACT

Retail banks usually measure growth by the number of new customers they acquire, but industry research shows that the number of products an existing customer holds is a stronger driver of long-term profitability and retention. This project examines low product penetration among retail banking customers, where a large share of the customer base holds only one product even though the bank offers several. Using a publicly available Kaggle dataset of 10,000 bank customers, it combines SQL-based analysis with a Power BI dashboard and the Key Influencers visual to identify which customer segments offer the highest-value cross-sell opportunities. The analysis finds that 50.84% of customers hold only one product, the average customer holds just 1.53 products, and these single-product customers together hold approximately $501.04 million in account balances. Geography had almost no influence on product penetration, while account balance was the strongest driver: customers with a zero balance were far less likely to be single-product (25.02%) than customers with any balance (64.52% to 70.59%). Power BI's Key Influencers feature confirmed that the Mid balance band ($100K–$200K) is the strongest predictor of single-product status, followed by the Low balance band and the 46–60 age group. The study concludes that banks should focus cross-sell campaigns on mid-to-low balance, single-product customers rather than targeting by geography or wealth alone, and that SQL and Power BI, without complex machine learning, are enough for banks of any size to identify and act on these opportunities.

TABLE OF CONTENTS

CH. No.	TOPIC	PG. No.
	Student Declaration	1
	Abstract	2
1	**Introduction**	4
2	**Objective of the Study** 2.1. Statement of the Problem 2.2. Objectives of the Study 2.3. Scope of the Study 2.4. Contribution of this Project	5-6
3	**Review of Literature**	7-9
4	**Research Methodology** 4.1. Research Design 4.2. Data Collection 4.3. Data Sources 4.4. Data Analysis	10-11
5	**Data Analysis and Interpretation**	12-15
6	**Result and Suggestions** 6.1. Findings of the Study 6.2. Suggestions	16-17
7	**Conclusion**	18
	Bibliography & References	19-20

CHAPTER 1

INTRODUCTION

In retail banking, customer growth is traditionally measured by the number of new accounts opened. However, a growing body of industry research shows that the depth of an existing customer relationship, specifically, how many products a single customer holds, is a more reliable predictor of long-term profitability than the size of the customer base alone. A customer who holds only one product, such as a savings account, typically remains with a bank for a much shorter period than a customer who holds two or three products, since additional products increase switching costs and deepen the customer's dependence on the institution.

This project examines the business area of customer product penetration and cross-sell analytics within retail banking. It investigates why a large proportion of bank customers remain single-product holders despite banks offering a wide range of financial products, and uses data analysis to identify which customer segments represent the most valuable, most actionable cross-sell opportunities. The study is grounded in a real, publicly available dataset of 10,000 retail banking customers, originally published on Kaggle, and applies SQL-based analysis together with Power BI dashboarding and built-in AI features to generate practical, data-driven recommendations.

The area is directly relevant to the field of business analytics, as it involves data cleaning, segmentation, aggregation using SQL, interactive dashboard design, and the interpretation of AI-generated insights, translating raw customer data into a clear business recommendation that a retail bank could act on.

CHAPTER 2

OBJECTIVE OF THE STUDY

2.1. Statement of the Problem:

Many retail banks have a large proportion of customers who hold only one product, such as a savings account, despite the availability of multiple products including credit cards, loans, and insurance. This represents a significant missed revenue opportunity, since multi-product customers are known to generate higher lifetime value and remain with the bank considerably longer than single-product customers.

Despite this, banks often lack a clear, segment-level view of which customers are most likely to be under-penetrated and which of these customers represent the highest-value opportunity. Without this visibility, cross-sell campaigns risk being poorly targeted, wasting marketing effort on low-probability segments while overlooking the customers most likely to convert. This study addresses this problem by using structured data analysis to identify exactly which customer segments carry the greatest untapped cross-sell potential.

2.2. Objectives of the Study:

- To analyze the overall level of product penetration among retail bank customers using SQL-based data analysis.
- To identify which customer attributes (geography, age, account balance, tenure) are associated with holding only a single product.

- To build an interactive Power BI dashboard, including an AI-driven Key Influencers analysis, to visualize and explain patterns in product penetration.
- To recommend a practical, data-driven cross-sell targeting strategy based on the findings.

2.3. Scope of the Study:

This study is based on a secondary dataset of 10,000 retail bank customers sourced from Kaggle. It covers customer-level attributes including geography, age, tenure, account balance, number of products held, and account activity status. The analysis is limited to the patterns observable within this dataset and does not involve primary data collection, customer surveys, or interviews with banking professionals.

2.4. Contribution of this Project:

This project contributes a practical demonstration of how SQL and Power BI, without the use of complex machine learning models, can be used to identify and prioritize cross-sell opportunities in retail banking. By combining structured query-based analysis with Power BI's built-in Key Influencers feature, the project shows that meaningful, explainable customer insights can be generated using tools that are accessible to analysts at banks of any size and technical maturity. The findings and methodology can serve as a reusable template for similar customer analytics exercises.

CHAPTER 3

REVIEW OF LITERATURE

1. Pillmore, B. (2026). "Cross-Selling Strategies in Banking: An Executive Blueprint."

This industry analysis argues that many banks measure growth using the wrong indicator, celebrating the number of new accounts added rather than identifying which existing households are under-monetized and could be served with more products. The author notes that this blind spot commonly arises because banks are organized around individual product lines, such as retail, mortgage, and wealth management, so no single team holds a complete view of the customer relationship.

2. Total Expert. (2026). "Banking Cross-Sell, Part 1: The Value of Engaged Customers."

This industry report finds that roughly a third of customers considering an additional financial product hesitate unless the bank proactively reaches out to them, suggesting that cross-sell opportunities often exist but go unrecognized and unacted upon by the institution.

3. Evok Advertising. (2026). "Elevate Your Bank's Cross-Selling Strategy."

This source presents industry data showing a strong relationship between product count and customer retention: a customer holding only one banking product typically stays with an institution for about eighteen months on average, compared to roughly four years for a two-product customer and close to seven years for a three-product customer.

4. Umbrex. (2026). "Customer Segmentation and Product Penetration Opportunities Analysis."

This retail banking industry framework recommends that banks segment their customer base by demographics, account information, and behavioral data, then assess how deeply each segment uses different product categories. It further recommends designing targeted product bundles for low-penetration segments rather than applying a single generic cross-sell strategy across the entire customer base, an approach this project directly applies through its balance-band and age-band segmentation.

5. Latinia. (2023/2026). "Facing the Data Silo Challenge: How Banks Can Overcome Fractured Customer Identities."

This source explains that banks are frequently structured around vertical product silos, meaning a customer's loan history, card usage, and account activity may sit in separate systems that do not communicate with one another, making it difficult to build a single, complete view of any one customer, a structural barrier to effective cross-selling.

6. Deloitte. (2026). "2026 Banking and Capital Markets Outlook."

Deloitte's industry outlook notes that many banks' data and analytics initiatives remain stuck as isolated pilot projects because the underlying customer data is fragmented and governed inconsistently across the organization, reinforcing those data fragmentation, not a lack of opportunity, is often the primary obstacle to effective cross-selling.

7. FI Works. (2026). "Why Banks and Credit Unions Can't Afford Data Silos in 2026."

This source adds a practical dimension to the data fragmentation problem, observing that in many banks, answering a simple business question about customer segments can take days or weeks because the process requires manually pulling data from several disconnected systems before any analysis can even begin.

8. Backbase. (2026). "How to Cross-Sell in Banking: Strategies, Examples, and AI-Driven Approaches."

This source notes that successful cross-selling depends as much on frontline staff understanding customer needs and on simple relationship-based incentives as it does on sophisticated analytics systems, suggesting that technology alone is not sufficient without an accompanying customer engagement strategy.

9. Forbytes. (2025/2026). "Cross-Selling in Banking: Strategies and Best Practices."

This industry analysis observes that while artificial intelligence and machine learning are increasingly used to identify cross-sell opportunities, the broader trend is toward better use of the customer data banks already collect, rather than adoption of complex new modeling techniques, directly supporting the SQL and Power BI-based approach used in this project.

10. Personetics. (2026). "Cross-Selling in Banking: Strategy, Examples & 2026 Trends."

This source notes that as more customers interact with their bank primarily through digital channels, engagement signals such as login frequency and transaction activity are becoming easier and more precise to observe, meaning segmentation analysis of the kind performed in this project is becoming more feasible over time, even for banks with limited analytics infrastructure.

Taken together, the literature points to a consistent conclusion: most banks already hold the customer data needed to identify under-penetrated, high-value segments, but this data is rarely organized or analyzed in a way that produces an actionable target list. This project's dashboard and findings directly address this gap using the customer-level dataset available for analysis.

CHAPTER 4

RESEARCH METHODOLOGY

4.1. Research Design:

This study adopts a quantitative, data-driven research design. Rather than relying on literature alone, the project applies direct analysis to a real customer-level dataset to identify measurable patterns in product penetration, and builds an interactive dashboard to communicate the findings.

4.2. Data Collection:

The dataset used for this study is the publicly available "Bank Customer Churn Prediction" dataset, sourced from Kaggle, containing 10,000 customer records. The dataset includes fields such as CustomerId, Geography, Age, Tenure, Balance, NumOfProducts, IsActiveMember, and EstimatedSalary. The NumOfProducts field, which ranges from 1 to 4, was repurposed in this study as a product-penetration metric; the dataset's original churn/Exited label was not used in this analysis.

Data Source (Kaggle): https://www.kaggle.com/datasets/saurabhbadole/bank-customer-churn-prediction-dataset

4.3. Data Sources and Tools:

The dataset was imported into a MySQL database, where all cleaning, segmentation, and aggregation queries were performed using SQL, including GROUP BY aggregation, CASE-based banding logic, and window functions. Power BI was then connected directly to the MySQL database to build a three-page interactive dashboard, incorporating calculated columns, DAX

measures, and Power BI's native Key Influencers AI visual to identify the strongest predictors of single-product status.

Code Repository (GitHub): https://github.com/PushpaR96/bank-product-penetration-analytics-using-sql-and-powerbi

4.4. Data Analysis:

Analysis was conducted in two stages. First, SQL queries were used to calculate product penetration rates, segment customers by geography, age band, and account balance band, and generate a ranked list of high-value, single-product customers. Second, Power BI's Key Influencers visual applied an automated statistical comparison between single-product and multi-product customers across all available attributes, ranking each attribute by the strength of its association with single-product status.

CHAPTER 5

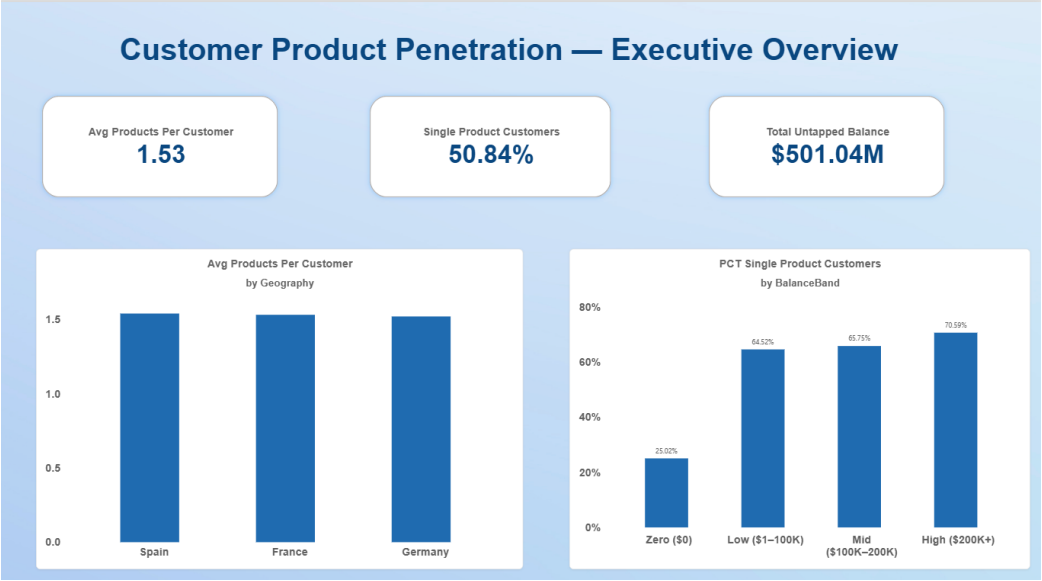
DATA ANALYSIS AND INTERPRETATION

1. Executive Overview — Product Penetration by Geography and Balance Band

Table No. 1

Metric	Value
Average products held per customer	1.53
Customers holding only 1 product	50.84%
Total balance held by single-product customers	$501.04 million

Chart No. 1: Customer Product Penetration — Executive Overview



Interpretation:

The dashboard shows that, on average, each of the 10,000 customers holds only 1.53 products, and just over half (50.84%) hold only a single product, representing $501.04 million in combined account balances. The left-hand chart shows average products held by geography (Spain, France, and Germany); all three countries sit within a narrow band (1.52–1.54), indicating that geography has almost no influence on product penetration and is not a useful variable for targeting cross-sell campaigns. The right-hand chart, showing the percentage of single-product customers by balance

band, reveals a far stronger and more counter-intuitive pattern: customers with a zero account balance are considerably less likely to be single-product (25.02%) than customers with any balance at all, whose single-product rate rises sharply to between 64.52% and 70.59% across the Low, Mid, and High balance bands. This suggests that having a non-zero balance does not, by itself, lead to additional product adoption, and that balance level, not geography, is the key variable for identifying cross-sell opportunity.

2. Opportunity Targeting — Identifying and Ranking Single-Product Customers

Table No. 2

Balance Band	Single-Product Customer Count
Zero ($0)	905
Low ($1–100K)	1,022
Mid ($100K–200K)	3,133
High ($200K+)	24

Chart No. 2: Cross-Sell Opportunity Targeting



Interpretation:

The left-hand table lists individual single-product customers, sorted by account balance in descending order, producing a ranked, actionable target list, for example, customer 15571958

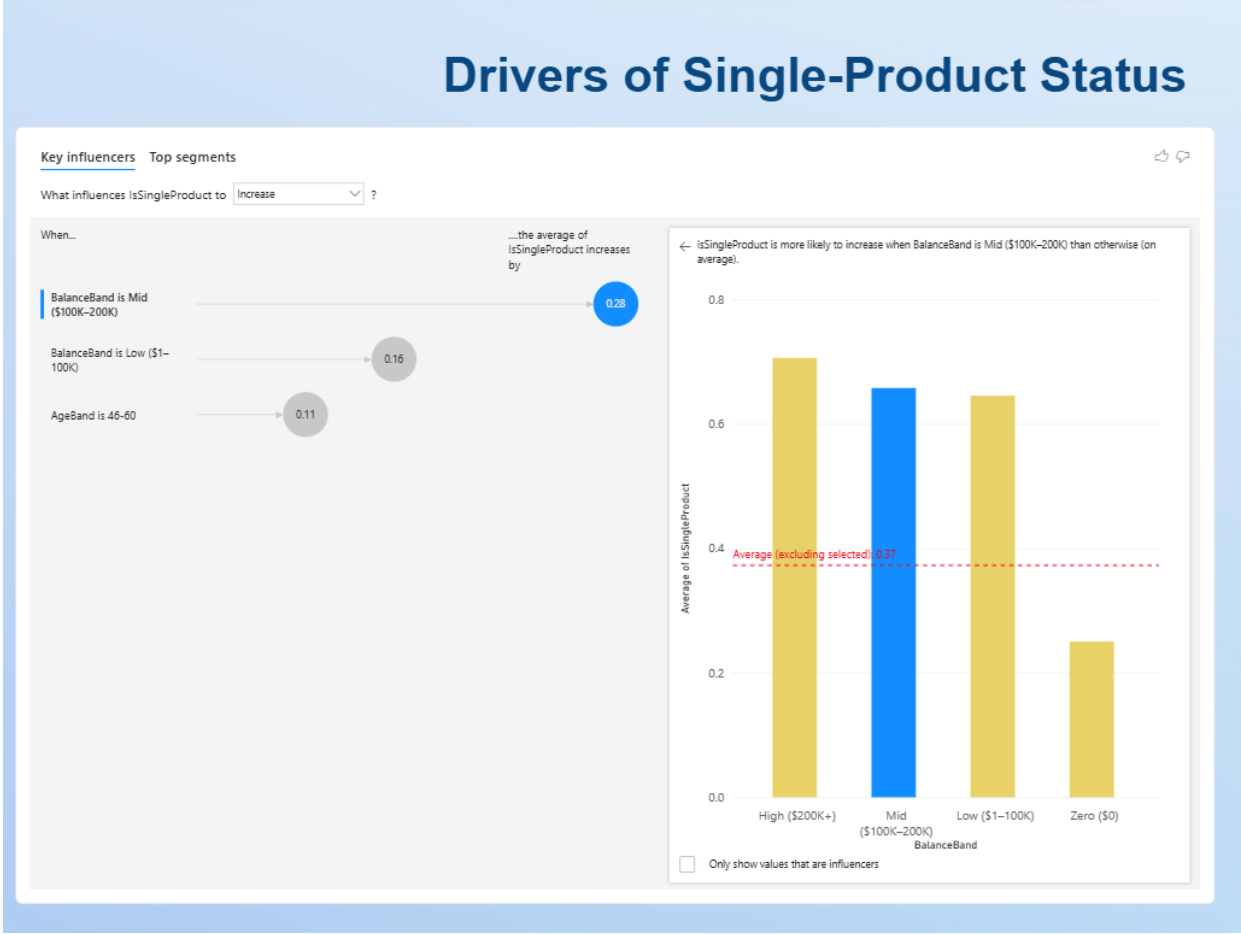
(Spain, age 40) holds a balance of $221,532.80 with only one product and ten similar high-balance customers appear at the top of the list. The right-hand bar chart shows that the largest concentration of single-product opportunities, by far, sits in the Mid balance band ($100K–$200K), with 3,133 customers, compared to only 1,022 in the Low band, 905 with a zero balance, and just 24 in the High band. This indicates that while high-balance customers individually represent the largest value per customer, the Mid balance band offers the greatest overall opportunity, combining a large customer count with meaningful individual balances, making it the most efficient segment for a bank to prioritize in a cross-sell campaign.

3. AI-Driven Insight — Key Influencers of Single-Product Status

Table No. 3

Influencing Factor	Increase in Likelihood of Single-Product Status
BalanceBand is Mid ($100K–200K)	+0.28
BalanceBand is Low ($1–100K)	+0.16
AgeBand is 46–60	+0.11

Chart No. 3: Drivers of Single-Product Status



Interpretation:

Power BI's Key Influencers visual independently confirms the pattern observed in the SQL-based analysis: being in the Mid balance band is the single strongest predictor of single-product status, increasing the average likelihood by 0.28 above the baseline average of 0.37. Being in the Low balance band is the second strongest predictor (+0.16), and belonging to the 46–60 age band is the third strongest (+0.11). No activity-status, tenure, or geography-based factor appeared among the top influencers, reinforcing the earlier finding that these variables carry little explanatory power for this outcome. The agreement between the independent SQL aggregation and the AI-generated Key Influencers output strengthens confidence that balance band, specifically the Mid band, is the genuine, primary driver of low product penetration in this customer base.

CHAPTER 6

RESULT AND SUGGESTIONS

6.1. Findings of the Study:

- Just over half of all customers (50.84%) hold only one banking product, and the average customer holds just 1.53 products overall, indicating a substantial, bank-wide cross-sell gap.
- Single-product customers collectively hold approximately $501.04 million in account balances, representing a significant pool of already-engaged, financially substantial customers who have not been sold additional products.
- Geography (Spain, France, Germany) has almost no influence on product penetration, with average products held varying only marginally (1.52 to 1.54) across all three countries.
- Account balance band is the strongest driver of single-product status: customers with a zero balance are far less likely to be single-product (25.02%) than customers with any balance at all (64.52% to 70.59%).
- The Mid balance band ($100K–$200K) contains by far the largest number of single-product opportunities (3,133 customers), compared to the Low (1,022), Zero (905), and High (24) bands.
- Power BI's Key Influencers analysis independently confirms that Mid balance band membership is the top predictor of single-product status (+0.28 above baseline), followed by Low balance band (+0.16) and the 46–60 age group (+0.11).

- Activity status, tenure, and geography did not emerge as meaningful predictors in either the SQL-based segmentation or the AI-driven Key Influencers analysis.

6.2. Suggestions:

- Banks should prioritize cross-sell campaigns toward customers in the Mid balance band, as this segment combines the largest customer count with meaningful individual account value, offering the best overall return on targeting effort.
- Geography should not be used as a segmentation variable for cross-sell targeting, given its negligible influence on product penetration in this analysis.
- Marketing and relationship teams should use ranked, balance-sorted target lists, similar to the one produced in this project, rather than broad, un-prioritized outreach to the entire single-product customer base.
- Banks should replicate this SQL plus Power BI analytical workflow on their own internal customer data, since it requires no complex machine learning and can be built and refreshed using the tools already available to most analytics teams.
- Further analysis could incorporate transaction-level or product-specific profitability data, where available, to refine the current balance-based segmentation into a more precise revenue-weighted targeting model.

CHAPTER 7

CONCLUSION

This study set out to identify which retail banking customer segments represent the greatest untapped cross-sell opportunity, using a combination of SQL-based analysis and an AI-assisted Power BI dashboard. The findings show that just over half of the customer base holds only one product, representing over half a billion dollars in collectively held balances, and that this opportunity is overwhelmingly concentrated in the Mid balance band rather than being evenly spread across geography or driven by account activity. The consistency between the SQL-derived segmentation and the independently generated Key Influencers output strengthens confidence in these conclusions. The project demonstrates that a bank does not require complex predictive modeling to identify high-value cross-sell opportunities; a disciplined, segment-based analysis using accessible tools such as SQL and Power BI is sufficient to produce a clear, actionable, and data-backed targeting strategy.

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